

PROGRAM CONTROL

RUN	Start; RUN "D:NAME"
LIST	LIST 10,50 / LIST "P: "
NEW	Erase program
CLR	Clear variables
CONT	Resume after STOP/BREAK
STOP / END	Halt program
SAVE / LOAD "D:N"	Tokenized program to disk
CSAVE / CLOAD	Cassette program
LIST / ENTER "D:N"	Save / merge as text
DOS	Enter the DOS menu
BYE	To Self-Test / Memo Pad

SAVE/LOAD = tokenized; LIST/ENTER = plain text you can merge.

BASIC STATEMENTS

PRINT ?	; joins, , tabs
INPUT	Read a line
LET	Assign (optional)
REM	Comment
GOTO GOSUB	Jump / call; RETURN
ON x GOTO	/ GOSUB
FOR .NEXT	STEP optional
IF .THEN	IF A>5 THEN 200
TRAP n	On error, GOTO line n
DATA READ	RESTORE rewinds
DIM	Required for arrays AND strings
POKE a,v	Write byte

No DEF FN. Multi-statement line with : .

FUNCTIONS

ABS SGN INT	magnitude/sign/floor
SQR EXP LOG CLOG	root, e^x, ln, log10
SIN COS ATN	DEG/RAD sets mode
RND(0)	random 0-1
PEEK(a)	read byte; POKE writes
FRE(0)	free bytes
LEN STR\$ VAL	string ↔ number
CHR\$ ASC	code ↔ char (ATASCII)
ADR(A\$)	memory address of string
USR(a,...)	call machine code

No **LEFT\$/MID\$/RIGHT\$** — take substrings with **A\$** (start,end).

GRAPHICS

ANTIC/GTIA

GRAPHICS n	Set mode (+16 full, +32 keep)
COLOR n	Pick colour/char to draw
SETCOLOR r,h,l	Register r → hue,lum
PLOT x,y	Plot a point
DRAWTO x,y	Line from last point
POSITION x,y	Move graphics cursor
LOCATE x,y,v	Read pixel into v

Fill region: **XIO 18,#6,0,0,"S:"** after plotting the outline.

GRAPHICS MODES

GR. n

0	40×24 text
1 / 2	20×24 / 20×12 big text
3	40×24 · 4 colour
5	80×48 · 4 colour
7	160×96 · 4 colour
8	320×192 · 1-colour hi-res

Add 16 for a full screen (no text window), 32 to keep current screen contents.

COLOUR REGISTERS

SETCOLOR

SETCOLOR 0-3	Playfield colours
SETCOLOR 4	Background / border
hue 0-15	Colour family
lum 0-14 even	Brightness
POKE 710,v	= SETCOLOR 2 (GR.0 bg)
POKE 712,v	Border colour
POKE 709,v	GR.0 text luminance

COLOR chooses which register a PLOT/DRAWTO uses.

SOUND

POKEY

SOUND v,p,d,vol	voice 0-3, pitch, distortion, volume
pitch 0-255	lower number = higher tone
distortion	0-14 even; 10 = pure tone
volume 0-15	0 = off
SOUND v,0,0,0	Silence one voice

Four independent voices. END or RUN resets all sound.

KEY PEEK / POKE

752	Cursor: POKE 752,1 hides it
710 / 712 / 709	bg / border / text lum
82 / 83	Left / right text margin
764	Last key (255 = none)
53279	Console keys: Start6 Sel5 Opt3
77	Attract: POKE 77,0 stops colour cycle
88 / 89	Screen RAM pointer (lo/hi)
20	jiffy clock (1/60 s)
16 & 53774	Disable the BREAK key

Controllers: STICK(0), STRIG(0), PADDLE(0), PTRIG(0).

DEVICES (CIO)

D: D1:-D8:	Disk drives
C:	Cassette
P:	Printer
E:	Screen editor
S:	Screen (graphics)
K:	Keyboard
R:	RS-232 (850 interface)
OPEN #c,m,a,"dev:n"	mode 4=read 8=write 12=update 6=dir
GET# PUT# / CLOSE#	Byte I/O; close channel
XIO cmd,#c,a1,a2,"d"	General device command

DOS & DISK

DOS 2.x

DOS	Enter the DOS menu from BASIC
A Directory	List files
C Copy / O Duplicate	Copy file / disk
D Delete E Rename	Manage files
I Format	Initialise a disk
L / K Binary	Load / save binary
B Run cartridge	Return to BASIC

Read a directory from BASIC: **OPEN #1,6,0,"D:*.":INPUT #1,A\$.**

EDITING & SPECIAL KEYS

BREAK	Stop a running program
SYSTEM RESET	Warm restart
CTRL + ↑↓←→	Move cursor
SHIFT+CLEAR	Clear screen (CTRL+<)
INSERT / DELETE	with CTRL = char, SHIFT = line
Atari key (♦)	Toggle inverse video
ESC then CTRL/key	Embed a control char in a string
CTRL+1	Pause / resume screen output

Hold OPTION/START/SELECT at boot for cartridge & diagnostic options.

QUICK-FACTS

CPU	6502 @ 1.79 MHz
Chips	ANTIC + GTIA video · POKEY sound/IO · PIA
400	8-16K, membrane keyboard
800	up to 48K, full keyboard, 2 cart slots
BASIC	OSS/Shepardson 8K cartridge
Error code ?	PEEK(195) after a TRAP